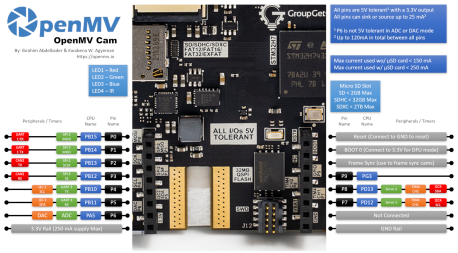
OpenMV® Pure Thermal

Datasheet

SKU: PURETHERMAL-OPENMV

Description

The PureThermal OpenMV is a co-designed collaboration between GetLab and OpenMV to produce a high-end mixed color and thermal camera for professional applications. Based on the OpenMV Cam H7 Plus platform, it combines a 5 MP color camera with a FLIR Lepton thermal sensor, a 4.3-inch touchscreen LCD, HDMI output, WiFi connectivity, and a laser rangefinder for thermal-to-color image alignment. Programmable in MicroPython via the OpenMV IDE with TensorFlow support for neural networks.



Target Areas

Thermal imaging, temperature measurement, industrial inspection, HVAC diagnostics, security, firefighting, wildlife monitoring, person detection, predictive maintenance.

Key Features

- **Dual imaging** — 5 MP OV5640 color camera with autofocus plus FLIR Lepton thermal sensor (supports Lepton 1/2/2.5/3/3.5). Per-pixel temperature measurement.
- **High-performance MCU** — STM32H743XI ARM Cortex-M7 at 480 MHz with 64 MB SDRAM and 32 MB QUADSPI flash.
- **Touchscreen display** — 4.3-inch 800x480 IPS TFT with 5-point capacitive touch and gesture support.
- **HDMI output** — Up to 1280x720 @ 60 Hz or 1920x1080 @ 30 Hz.
- **WiFi** — WINC1500 SoC for MJPEG streaming via RTSP at up to 12 Mb/s or general internet connectivity.
- **Laser rangefinder** — VL53L0X time-of-flight sensor, measures up to 2 m for thermal-to-color alignment.
- **MicroPython + TensorFlow** — Program in Python via OpenMV IDE with TensorFlow support for neural networks.
- **Rich I/O** — 10 GPIO pins with SPI, I2C, UART, CAN, ADC, DAC, PWM, and 2 servo channels. USB-C and Qwiic connector.

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1 Features and System Components

1.1 Microcontroller

Component	Details
STM32H743XI	ARM Cortex-M7 core running at 480 MHz. Features double-precision floating-point unit (FPU), DSP instructions, and hardware JPEG codec.

1.2 Memory

Component	Details
External SDRAM	64 MB SDRAM at 400 MB/s bandwidth.
QUADSPI Flash	32 MB QUADSPI flash at 50 MB/s for program and data storage.
microSD Card	Supports SD/SDHC/SDXC cards. FAT12/FAT16/FAT32/exFAT file systems.
RTC	Battery-backed real-time clock for date/time stamping.

1.3 Connectivity

Component	Details
USB-C	USB 2.0 Full-Speed (12 Mbps). Power input, battery charging, and PC data connection. Virtual COM port and mass storage access to QUADSPI flash and microSD.
WiFi	WINC1500 SoC (FCC certified). MJPEG streaming via RTSP at up to 12 Mb/s or general internet connectivity.
HDMI	HDMI output up to 1280x720 @ 60 Hz or 1920x1080 @ 30 Hz (dynamic) / 60 Hz (static).
Qwiic / I2C	SparkFun Qwiic connector (I2C). Compatible with Seeed Grove and Adafruit STEMMA.
SPI	1 SPI bus on GPIO header, up to 80 Mb/s.
UART	2 asynchronous serial (UART) ports on GPIO header. Up to 7.5 Mb/s.
CAN	1 CAN bus on GPIO header. Up to 1 Mb/s.

1.4 Color Camera

Component	Details
Image Sensor	OV5640 5 MP CMOS image sensor (2592x1944) with autofocus support.
Lens	Integrated lens with autofocus.

1.5 Thermal Sensor

Component	Details
FLIR Lepton Socket	Supports FLIR Lepton 1, 2, 2.5, 3, and 3.5 modules (both radiometric and non-radiometric). 80x60 or 160x120 pixel thermal resolution depending on Lepton version.
Thermal Performance	Long-wave infrared (LWIR) imaging, 8–14 µm spectral range. Per-pixel temperature measurement in Celsius (radiometric Lepton models). NETD < 50 mK.
Frame Rate	Up to 8.7 Hz thermal frame rate.

1.6 Laser Rangefinder

Component	Details
VL53L0X	Time-of-flight laser rangefinder. Measures up to 2 m distance. Used for thermal-to-color image alignment offset and scaling.

1.7 Display

Component	Details
LCD	4.3-inch 800x480 IPS TFT display. 24-bit color, 60 Hz refresh rate. Backlight dimming.
Touch	Capacitive touch with up to 5-point multi-touch. Gesture detection: pinch zoom in/out, swipe left/right/up/down.

1.8 Other Components

Component	Details
RGB LED	User-programmable RGB LED indicator.
White LED	High-brightness white LED illuminator.
Buzzer	Programmable buzzer with volume and frequency control.
Debug	Standard ARM SWD-JTAG debug connector.

1.9 Power Management

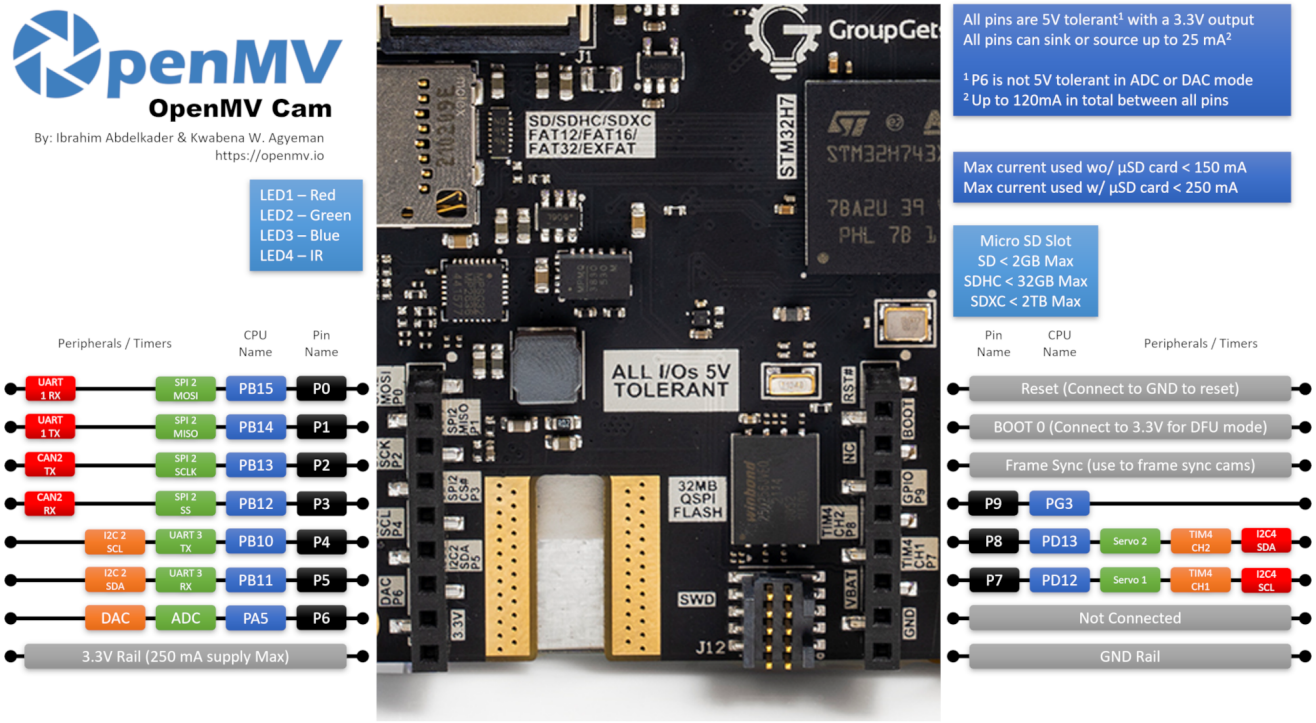
Component	Details
USB-C	Power input and LiPo battery charging via USB-C connector.
LiPo Battery	1S LiPo battery connector. Battery charges automatically when USB-C is connected.

2 Ratings

2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
VUSB	USB-C supply voltage	4.5	5.0	5.5	V
VBATT	LiPo battery voltage	3.0	3.7	4.2	V
TOP	Operating temperature	-10	25	65	°C

3 Connector Overview



3.1 GPIO Header (10 pins)

Pin	Label	Type	Alternate Functions
1	P0	I/O	UART RX / SPI MOSI
2	P1	I/O	UART TX / SPI MISO
3	P2	I/O	CAN TX / SPI SCLK
4	P3	I/O	CAN RX / SPI SS
5	P4	I/O	I2C SCL / UART TX
6	P5	I/O	I2C SDA / UART RX
7	P6	I/O	DAC / ADC — 3.3V tolerant
8	P7	I/O	Servo1 / PWM
9	P8	I/O	Servo2 / PWM
10	3V3	Power	3.3 V regulated output

4 Functional Overview

4.1 Power Tree

The PureThermal OpenMV is powered via USB-C or a 1S LiPo battery. The USB-C port provides both power and data connectivity. When USB is connected, the integrated charger charges the LiPo battery automatically.

5 General Info

Dimension	Value
Weight	121 g
Operating Temperature	-10°C to +65°C
Storage Temperature	-40°C to +80°C

6 Certifications

Certification	Region	Details
RoHS	EU	Compliant with EU Directive 2011/65/EU (RoHS 2). All components and solder are lead-free.
REACH	EU	Compliant with EU Regulation (EC) No 1907/2006.

7 Company Information

Field	Details
Co-designed by	GetLab / GroupGets
Website	https://openmv.io
Documentation	https://docs.openmv.io
Support	https://forums.openmv.io

8 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/openmv-pure-thermal
Documentation	https://docs.openmv.io
Community Forum	https://forums.openmv.io
FLIR Lepton	https://www.flir.com/products/lepton/
OpenMV IDE	https://openmv.io/pages/download

9 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.